

LAB 3

COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Problem Statement #09

FACULTY RESEARCH GRANT & PUBLICATION TRACKER

Name: J Karthik	SRN: PES1UG25AM804	Section: B
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1. Objective

Evaluate architectural styles, select the most appropriate architecture for the Faculty Research Grant & Publication Tracker, and document the architectural decision.

2. Architectural Style Analysis

Layered Architecture: separates the system into Presentation, Business, and Data layers. It provides clear separation of concerns and is suitable for a structured portal application.

Microservices Architecture: provides independently deployable services and independent scaling, but introduces operational complexity and data-consistency challenges.

Client-Server Architecture: provides centralized control and simple deployment, but can create a scalability bottleneck and a single point of failure.

3. Architecture Selection

We chose Layered Architecture for the Faculty Research Grant & Publication Tracker.

4. Components Identified

Web Portal UI – Faculty Researcher screens and Research Dean dashboard.

- Authentication & Access – authentication and role-based access.
- Grant & Expense Manager – grant records, expenses, and expense-cap validation.
- Publication Manager – publications, indexing status, and citations.
- Approval Workflow – co-author approval and approval-status tracking.
- Analytics Service – fund burn-up and research/publication performance metrics.
- Database Repository – grants, expenses, publications, and users.
- Audit Ledger – immutable, tamper-evident audit records.

5. Interfaces

IGrantService – grant management.

- IPublicationService – publication management.
- IApprovalReview – approval review.
- IAnalyticsService – analytics and reporting.
- IApprovalRequest – internal approval request.
- IDataAccess – parameterised database access.
- IAuditLog – append-only audit logging.

6. Written Justification

Reason 1 – Separation of Concerns: Grant management, expense validation, publication management, approval workflow, and analytics are separated into focused business components, making the system easier to maintain.

Reason 2 – Clear Component Interaction: The portal communicates with business services through defined interfaces, while business components access the database and audit ledger through dedicated interfaces.

Security Advantage: The Immutable Audit Ledger provides a tamper-evident record of financial approvals and publication-status modifications. Authentication and role-based access protect system operations.

Performance Benefit: Separating presentation, business processing, and data access keeps responsibilities organized and supports responsive access during expected peak usage.

7. Requirements Covered

FR-001: Grant records and approved funding limits.

FR-002: Research expenses, balance calculation and expense-limit validation.

FR-003: Publication and citation/indexing records.

FR-004: Co-author approval workflow.

FR-005: Fund burn-up and research/publication analytics.

NFR-001: Immutable, tamper-evident audit ledger.

NFR-002: Responsive access under expected peak usage.